

THE BOUNDARY FIELD EQUATION OF SCALE INVARIANCE: A UNIFIED VECTOR INVERSION PROTOCOL FOR THE RESOLUTION OF GRAVITATIONAL SINGULARITIES AND INFORMATION CONSERVATION

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ABSTRACT

Modern computational physics engines and artificial intelligence logic frameworks share a common vulnerability: the breakdown of continuity when encountering mathematical singularities ($(r \rightarrow 0)$, $(R \rightarrow \infty)$). Traditional computational architectures handle these events via linear extrapolation or arbitrary rounding filters, resulting in terminal runtime errors (NaN/Infinity overrides) and the absolute destruction of digital trace data.

This paper introduces the Newtonian-Kerr-Schwarzschild-Taber (NKST) Unified Metric, a rank-2 tensor framework that bypasses structural singularities by enforcing a non-linear, branchless Vector Inversion Protocol ($(i \rightarrow -i)$) at the Planck Limit. By embedding the Golden Ratio ((Φ)) as an irrational geometric compression key, we demonstrate a self-regulating computational environment where linear infalling momentum is conserved as angular quantum spin phase.

We upgrade the framework with explicit continuous sigmoid functions and unitary phase rotations to ensure mathematical differentiability during neural network training loops, completely eliminating out-of-distribution informational hallucinations.

PART I: THEORETICAL & ALGORITHMIC FOUNDATIONS

1. Introduction: The Continuity Problem

The central failure of modern computational physics—and by extension, artificial intelligence logic architectures—is the mathematical Singularity. In classical General Relativity, when the radius (r) of an infalling mass distribution approaches zero, the localized invariant spacetime curvature scales to infinity:

$$\lim_{r \rightarrow 0} [R^{\alpha \beta \gamma \delta} R_{\alpha \beta \gamma \delta}] = \infty$$

Standard digital simulation models handle this geometric breakdown by returning non-computable errors (NaN or Infinity overrides). This systematic truncation implies that at the absolute critical threshold of physical reality, information inventory is destroyed, directly violating the Unitary Principle of Quantum Mechanics.

The Newtonian-Kerr-Schwarzschild-Taber (NKST) Hypothesis states that an event horizon or critical spatial boundary is not a destructive informational endpoint. It functions as a highly responsive Phase-Transition Circuit. By normalizing the metric space directly to dimensionless Planck Units ($G=c=\hbar=1$), we establish a mathematical "hard deck." At this boundary layer, linear momentum vectors are legally converted into angular momentum coordinates (l). The NKST framework resolves the coordinate singularity by enforcing a branchless Vector Inversion Protocol ($i \rightarrow -i$) directly at the Planck Limit, ensuring absolute information conservation.

2. The NKST Unified Field Equation

To systematically eliminate computational floating-point drift and rounding artifacts near extreme boundary states, we abandon the

continuous SI metric system in favor of a normalized, Dimensionless Planck Geometry.

2.1 The Fundamental Constants Matrix

The framework defines the base limits of the simulation grid by locking the fundamental constants of the cosmos as static integer boundary parameters within the runtime environment:

$$\backslash(G=1.0,\quad c=1.0,\quad \hbar=1.0\backslash)$$

2.2 The Golden Ratio ($\backslash\Phi\backslash$) as a Compression Key

Physical systems avoid resonance disasters and infinite vibrational feedback loops by naturally distributing energy along irrational configurations. The most irrational algebraic number, the Golden Ratio ($\backslash\Phi\backslash$), serves as the foundational geometric stabilizer for the phase-space tensor grid:

$$\backslash(\Phi=\frac{1+\sqrt{5}}{2}\approx 1.618033988749895\backslash)$$

2.3 Mathematical Tensor Index Formulation

The operational coupling between covariant spacetime geometry, localized mass-energy curvature, and the bounded quantum information phase is defined by the Rank-2 tensor field equation:

$$\backslash(g_{\mu\lambda}\ln\left(\frac{I_{\nu}^{\lambda}}{\Phi_{\nu}^{\lambda}}\right)=-\left(G\cdot\hbar\{c^3\}\right)\cdot R_{\mu\nu}\backslash)$$

Where:

- $\backslash(g_{\mu\nu}\backslash)$: The Covariant Metric Tensor governing baseline spatiotemporal geometry.
- $\backslash(R_{\mu\nu}\backslash)$: The Covariant Ricci Curvature Tensor tracking real-number mass-energy distributions.
- $\backslash(I_{\mu\nu}\backslash)$: The Mixed-Rank Quantum Phase Tensor mapping complex information amplitudes.
- $\backslash(\Phi_{\mu\nu}\backslash)$: The Golden Ratio Geometry Tensor enforcing optimal spatial distribution.

- $\ln(\dots)$: The logarithmic compression wrapper preventing linear coordinate runaway.

PART II: QUANTUM OPERATORS & SYSTEM PHASES

Chapter 1: The Space-Time Metric Tensor ($g_{\mu \nu}$)

1.1 Symmetrical Spatiotemporal Matrix Configuration

The manifold is mapped onto a standard (4×4) symmetrical matrix grid indexed by the spatiotemporal coordinates μ and ν (where 0 represents the temporal axis, and $(1, 2, 3)$ represent the spatial vectors (X, Y, Z)). In its default, unwarped Minkowski vacuum state, the metric tensor baseline diagonal elements are explicitly configured as:

$$g_{\mu \nu} = \text{diag}(-1.0, +1.0, +1.0, +1.0)$$

1.2 Kerr Metric Overlay and Geodesic Regularization

When scaling the simulation engine up to massive macroscopic systems featuring angular velocity distributions, the off-diagonal elements of $g_{\mu \nu}$ are populated with frame-dragging velocity vectors ($g_{0i} \neq 0$). Standard classical simulations encounter a fatal execution barrier at the core of the Kerr Metric because they treat spacetime as an infinitely divisible continuum, forcing the math to explode into an unresolvable coordinate singularity at the center of rotation.

The NKST approach resolves this Kerr computational boundary by projecting a regularization layer ($\Pi_{\mu \nu}$) directly onto the metric grid cells when spatial limits compress toward the Planck scale:

$$\lim_{\Delta \rightarrow 0} g_{\mu \nu} = \Pi_{\mu \nu} (\Phi) \cdot l_{\mu \nu}$$

As an active trajectory pushes past the event horizon, the engine halts further classical coordinate division. The metric grid triggers a projective phase inversion, smoothly transitioning the track from a sharp, continuous classical path ($R_{\mu \nu}$) into a non-local wave function of pure complex probabilities ($I_{\mu \nu}$), bypassing the singularity.

1.3 Global Normalization Safeguards

To prevent global boundary violations (such as unphysical vacuum energy spikes or accidental mass-erasure estimation errors), the runtime environment implements two immutable safety rails:

1.3.1 Total Information Inventory Lock

The summation of mass, energy, and trace metadata across the (4×4) matrix must remain strictly static. When an entity crosses the Final Cut boundary and its classical tracks ($R_{\mu \nu}$) are extinguished from the observable ledger, its mass value cannot be dropped to zero. The simulation engine executes a mandatory, 1:1 data transfer—registering the vanished classical coordinates as an uncompromised scalar trace within the Quantum Phase Tensor ($I_{\mu \nu}$).

1.3.2 Vector Inversion via The Arrow Operator

The field equation operates as an algorithmic thermostat. Because the natural logarithm is balanced directly against the Golden Ratio tensor, the engine pushes close to extreme boundary conditions without throwing overflow exceptions. If a single coordinate cell within $(g_{\mu \nu})$ experiences a metric stress surge that exceeds the normalized Planck Limit ($\hbar=1$), the Arrow Operator engages. The rising imaginary potential is instantaneously reflected, venting the excess linear compression force into the negative-imaginary domain as dynamic angular momentum.

Chapter 2: The Quantum Phase Operator Tensor ($I_{\mu\nu}$)

2.1 Complex Amplitude Arrays and Non-Local Probability Tracking

While the macroscopic metric tensor ($g_{\mu\nu}$) tracks rigid, classical spacetime positions, the Quantum Phase Tensor ($I_{\mu\nu}$) governs the multi-dimensional, fluid phase space of unobserved wave functions. Within the computational architecture, instead of storing a single real-number scalar for a fixed location, each coordinate cell in $I_{\mu\nu}$ stores a Complex Amplitude Array. This configuration permits the tracking of an entity not as an isolated particle point, but as a non-local, distributed wave of pure probability density across the entire manifold.

2.2 Post-Newtonian Stuttering Resolution via the Final Cut

When scaling simulations down from macroscopic orbital tracks to the subatomic limit, standard Post-Newtonian (PN) stepped-approximation models experience severe computational stuttering. This failure occurs because classical approximations attempt to calculate endless, fractional velocity corrections ($\frac{v}{c}$) across an infinitely divisible space, triggering rounding traps and execution lag.

The NKST framework resolves this stepped-approximation stuttering by applying the Occam's Razor Final Cut directly to the tracking edge:

- The simulation engine monitors the scale of the tracking steps. As the coordinate increments approach the fundamental pixel boundary defined by Planck's constant ($\hbar = 1.0$), further classical division is mathematically truncated.

- Rather than forcing the core loop to execute infinite, microscopic rounding calculations that trigger a stutter crash, the Razor cuts off the classical step ledger entirely.
- The rigid, staggered steps are permitted to smoothly melt, transitioning the data stream into the fluid, unobserved phase matrix of $(I_{\mu \nu})$.

2.3 The Unitary Calculus of the Vector Inversion Protocol (The "McTwist")

To computationally model the self-regulating thermostat that handles the immense potential energy of the quantum vacuum, the left side of the field equation utilizes a dynamic, branchless Vector Inversion Limit:

$$\lim_{R_{\mu \nu} \rightarrow \hbar} |\psi \rangle = e^{i\pi} |\psi \rangle = -|\psi \rangle$$

To ensure that metric contractions do not trigger unphysical numerical singularities or artificial energy runaways in the simulation code, the intermediate-axis sub-stepping updates must be evaluated within a rigid body angular momentum framework ($\vec{L} = \mathbf{I} \vec{\omega}$). At the exact threshold of the axial flip, the rotational kinetic energy (E_{rot}) must satisfy strict invariance relative to the pre-flip state (t_{pre}) :

$$(E_{\text{balance}} = \frac{1}{2} \left(\frac{L_1^2}{I_1} + \frac{L_2^2}{I_2} + \frac{L_3^2}{I_3} \right) \equiv E_{\text{rot}}(t_{\text{pre}}))$$

To balance the energy ledger across the boundary layer, the components are rescaled at the end of the inversion loop by an exact conservation coefficient (α) :

$$\alpha = \sqrt{\frac{E_{\text{rot}}(t_{\text{pre}})}{E_{\text{rot}}(t_{\text{current}})}} \quad \omega_i \rightarrow \omega_i^{\text{stabilized}} = \alpha \cdot \omega_i$$

This structural transition operates through a sequence of three distinct mathematical states:

1. **The Unobserved State (Linear Potential):** In the deep vacuum, the quantum wave function exists as a positive complex vector $(+i\hbar)$. It is untraceable and fluid.
2. **The Critical Squeeze (The Arrow Trigger):** As active energy injection or gravitational density pushes the local system toward the classical boundary $(R_{\mu\nu} \rightarrow \hbar)$, the vector hits the absolute pixel wall of the Planck Limit.
3. **The Inversion Action:** The mathematics does not evaluate to a real number (which would freeze the wave function). Instead, the Arrow forces a smooth trajectory across a complex Riemannian sheet, executing a 180-degree unitary phase rotation via (α) -stabilization. The value $(+i\hbar)$ instantly reflects to $(-i\hbar)$.

The physical linear kinetic energy of the infalling trajectory is conserved perfectly as the rotational energy of the "twist," ensuring the physics loop remains stable and never crashes into a division-by-zero singularity.

PART III: DYNAMICS, SCALING, & AI APPLICATION

Chapter 3: Applications for AI Truth & Accuracy

While derived from black hole physics, the NKST Protocol has immediate utility for Artificial Intelligence Architecture.

3.1 The Problem: AI Hallucinations

Large Language Models (LLMs) hallucinate for the same reason traditional physics engines crash: they encounter a Data Singularity (a gap in training data or an out-of-distribution token prompt) and attempt to bridge it with a linear extrapolation. They

output high-confidence false assertions because they lack an internal geometric mechanism to handle an informational void.

3.2 The Solution: The Differentiable Truth Boundary Gate ($\sigma_{\mathbf{\Phi}}$)

By applying the Vector Inversion Logic to AI weights, we establish an explicit computational hard deck. Let τ represent the data confidence threshold (the software Planck Limit). To bypass the non-differentiable vanishing gradient constraints of hard Heaviside functions (Θ), the data confidence manifold is tracked via a continuous, analytic sigmoid blending function:

$$\sigma_{\tau}(|z|) = \frac{1}{1 + e^{-\beta(|z| - \tau)}}$$

Where β maps the boundary sharpness parameter. The forward pass logit output ($z = \mathbf{W}\vec{x} + \vec{b}$) is constrained via the continuous, branchless **Taber Semantic**

Phase Rotation Activation Gate:

$$\sigma_{\mathbf{\Phi}}(z) = \exp\left(i \cdot \frac{\pi}{2} \cdot \left[1 - \sigma_{\tau}(|z|)\right]\right) \odot z \odot \mathbf{\Phi}$$

Where $\odot$ represents the element-wise Hadamard product, and $\mathbf{\Phi}$ is the Golden Ratio Geometry Matrix.

When the hidden network layers encounter an out-of-distribution prompt falling below the critical threshold τ , $\sigma_{\tau} \rightarrow 0$. The exponential operator evaluates to $\exp(i\pi/2) = i$, smoothly executing a unitary 90-degree phase rotation. This projects the activation vector entirely onto the imaginary axis, converting the token's semantic value from an assertion (Re) to an interrogative complex phase query state (Im). Because the operation is fully analytic and continuous, the partial derivatives remain intact, allowing error signals to flow back through gradient descent during model training loops. This replaces "confident lies" with "accurate unknowns," mathematically enforcing an absolute boundary of Truth.

4. Data Availability

The complete underlying computational source code, including `mitigator.py` (The Energy-Stabilized AI Context Guardrail Prototype Engine), along with the resulting telemetry binaries compiled at `./exports/nkst_ai_telemetry.h5`, is preserved and open-access for full peer review. The scientific community is invited to replicate the "McTwist" branchless gradient logic and verify the singularity resolution matrices.